

REPORT

Digital Image Enhancement, Restoration and Object Segmentation Using the Oxford-IIIT Pet Dataset

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1.Title

Digital Image Enhancement, Restoration and Object Segmentation Using the Oxford-IIIT Pet Dataset

2. Problem Statement

Digital images may contain noise, poor contrast, unclear details, and indistinct object boundaries. These problems can affect the quality and interpretation of images.

This project focuses on processing pet images to improve their visual quality, add and remove different types of noise, detect object boundaries, and separate the pet from the background. Spatial image processing, filtering, thresholding, edge detection, and morphological techniques are used to achieve these objectives.

3. Objectives

The main objectives of this project are:

- To preprocess pet images for image processing.
- To improve image quality using enhancement techniques.
- To study Gaussian and salt-and-pepper noise.
- To remove noise using mean, median, and Wiener filtering.

- To detect image boundaries using Sobel and Canny edge detection.
- To segment the pet using Otsu thresholding.
- To improve segmentation using morphological operations.
- To compare the segmentation result with the ground-truth mask.
- To evaluate restoration using MSE, PSNR, and SSIM.
- To evaluate segmentation using IoU, Dice Score, and Accuracy.

4. Dataset / Image Description

The **Oxford-IIIT Pet Dataset** is used for this project. It contains images of different breeds of cats and dogs along with segmentation annotations.

The dataset is loaded using the OxfordIIITPet dataset from Torchvision with the trainval split and segmentation masks.

The selected images are resized to **256 × 256 pixels** for processing. The RGB images are converted into grayscale images for intensity-based enhancement, noise filtering, thresholding, and edge detection.

The segmentation masks are resized using nearest-neighbor interpolation to preserve the mask labels. These masks are used as ground truth for evaluating the segmentation results.

5. Methodology

The project follows the following workflow:

Input Image → Preprocessing → Image Enhancement → Noise Addition → Noise Removal / Restoration → Edge Detection → Segmentation → Morphological Post-processing → Ground Truth Comparison → Evaluation

Image Enhancement

The following techniques are applied:

- Gamma Transformation
- Contrast Stretching
- Histogram Equalization

These techniques are used to improve image brightness, contrast, and intensity distribution.

Noise Addition and Restoration

Two types of noise are introduced:

- Gaussian Noise
- Salt-and-Pepper Noise

The noisy images are restored using:

- Mean Filtering
- Median Filtering
- Wiener Filtering

Segmentation

The following techniques are used:

- Sobel Edge Detection
- Canny Edge Detection
- Otsu Thresholding

Morphological Processing

The following morphological operations are applied:

- Erosion
- Dilation
- Opening
- Closing

These operations are used to process and improve the binary segmentation result.

6. Implementation

The project is implemented using Python.

The main libraries used are **NumPy**, **OpenCV**, **Matplotlib**, **Pandas**, **Torchvision**, **SciPy**, and **scikit-image**.

The implementation includes dataset loading, preprocessing, image enhancement, noise generation, noise restoration, edge detection, Otsu thresholding, morphological processing, ground-truth comparison, and evaluation.

The restoration results are evaluated using **Mean Squared Error (MSE)**, **Peak Signal-to-Noise Ratio (PSNR)**, and **Structural Similarity Index (SSIM)**.

The segmentation results are evaluated using **Intersection over Union (IoU)**, **Dice Score**, and **Accuracy**.

7. Results

The project produces visual and quantitative results for image enhancement, restoration, and segmentation.

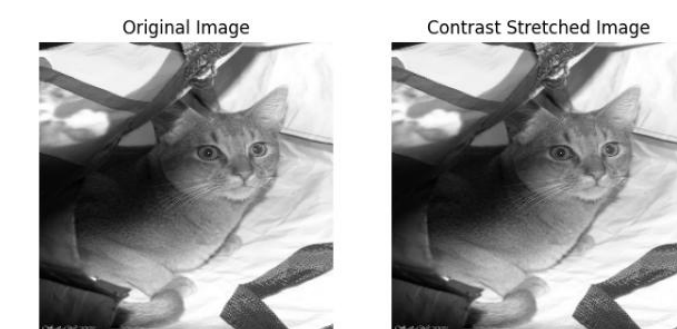
7.1 Image Enhancement Results

Gamma Transformation, Contrast Stretching, and Histogram Equalization are applied to the grayscale image. These techniques improve the brightness, contrast, and intensity distribution of the image.

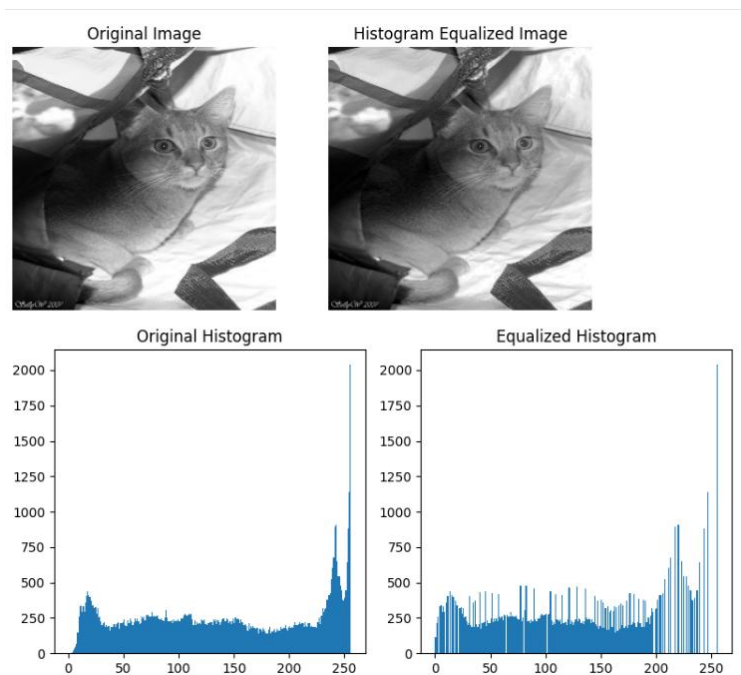
Gamma Transformation Result



Contrast Stretching Result



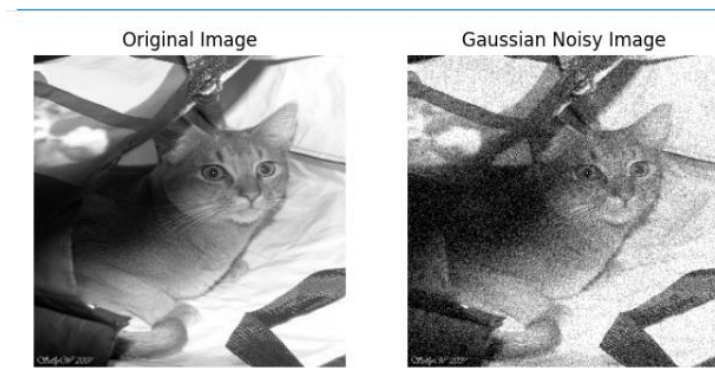
Histogram Equalization Result



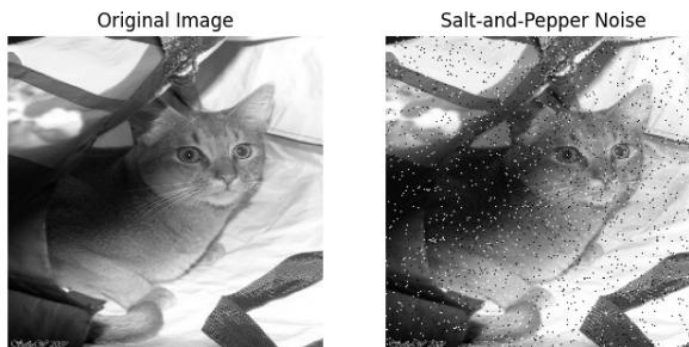
7.2 Noise Addition Results

Gaussian noise and salt-and-pepper noise are added to the grayscale image.

Gaussian Noise Result



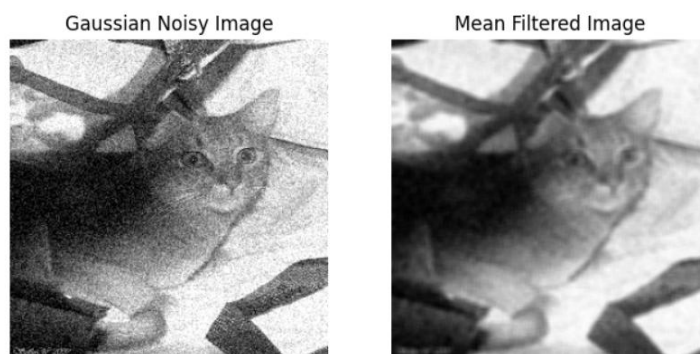
Salt-and-Pepper Noise Result



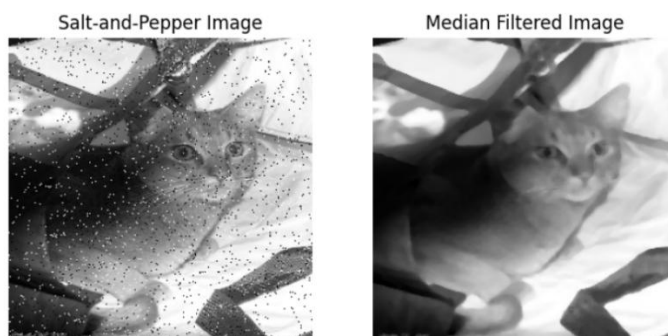
7.3 Noise Restoration Results

Mean filtering is applied to the Gaussian noisy image, while median filtering is applied to the salt-and-pepper noisy image. Wiener filtering is also applied to the Gaussian noisy image.

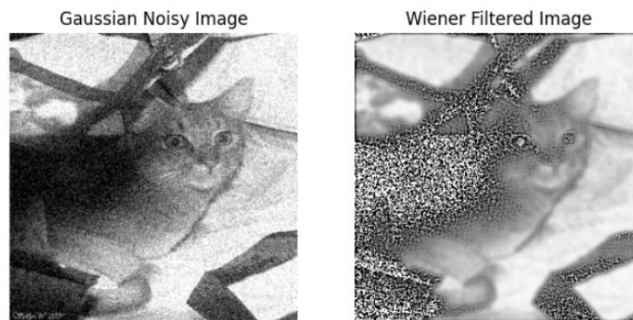
Mean Filtering Result



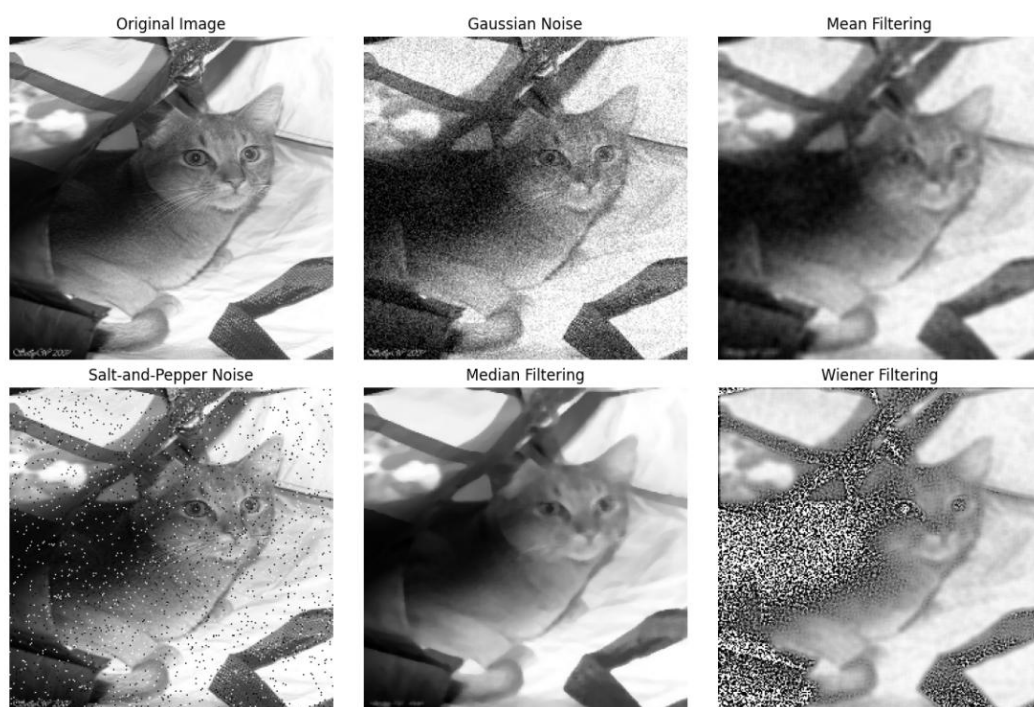
Median Filtering Result



Wiener Filtering Result



Restoration Results



The restoration techniques are evaluated using **MSE, PSNR, and SSIM**. The actual values obtained from the program execution should be included in the report.

Restoration Evaluation

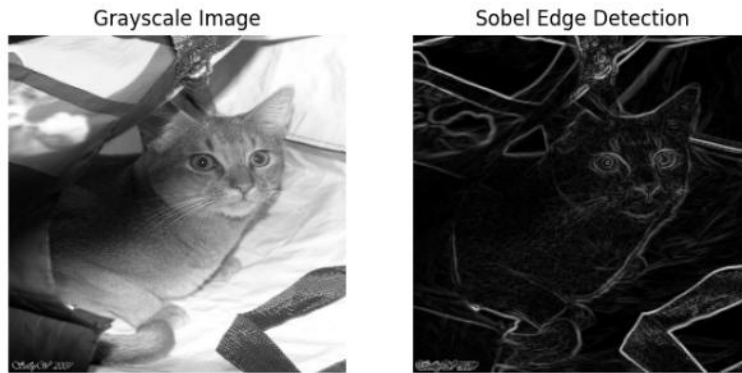
The restoration techniques are evaluated using MSE, PSNR, and SSIM.

```
Mean Filtering
MSE : 204.16783142089844
PSNR: 25.030930449433647
SSIM: 0.7169416036803673
Median Filtering
MSE : 117.0440673828125
PSNR: 27.447309553786617
SSIM: 0.8217245018444234
Wiener Filtering
MSE : 6218.71435546875
PSNR: 10.193797520673716
SSIM: 0.3546072507941955
```

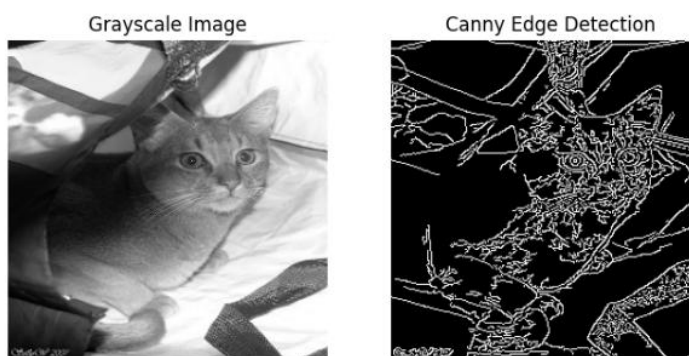
7.4 Edge Detection Results

Sobel and Canny edge detection are applied to identify important boundaries in the pet images.

Sobel Edge Detection Result



Canny Edge Detection Result



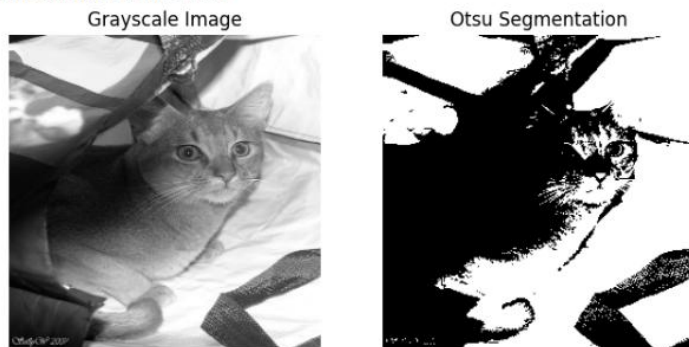
7.5 Segmentation Results

Otsu thresholding is used to automatically create a binary segmentation mask.

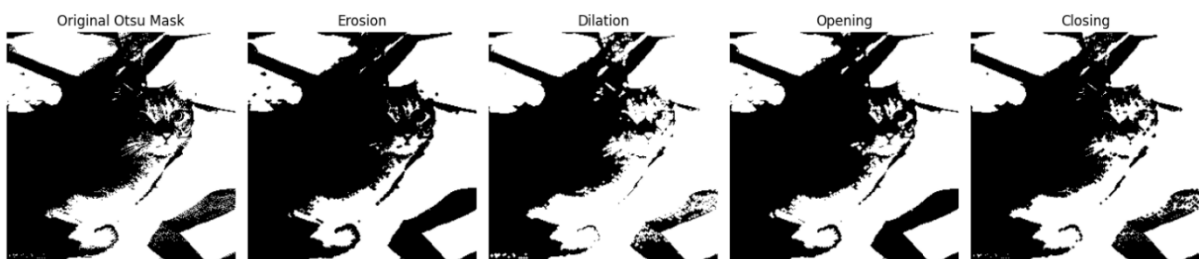
Morphological operations are then applied to improve the segmentation result.

Otsu Segmentation Result

Otsu threshold value: 143.0



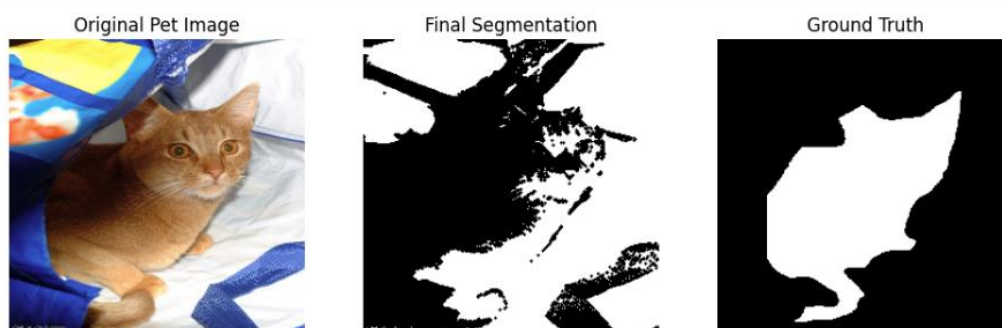
Morphological Operations Result



7.6 Ground Truth Comparison

The final segmentation result is compared with the ground-truth mask provided by the Oxford-IIIT Pet Dataset.

Ground Truth Comparison

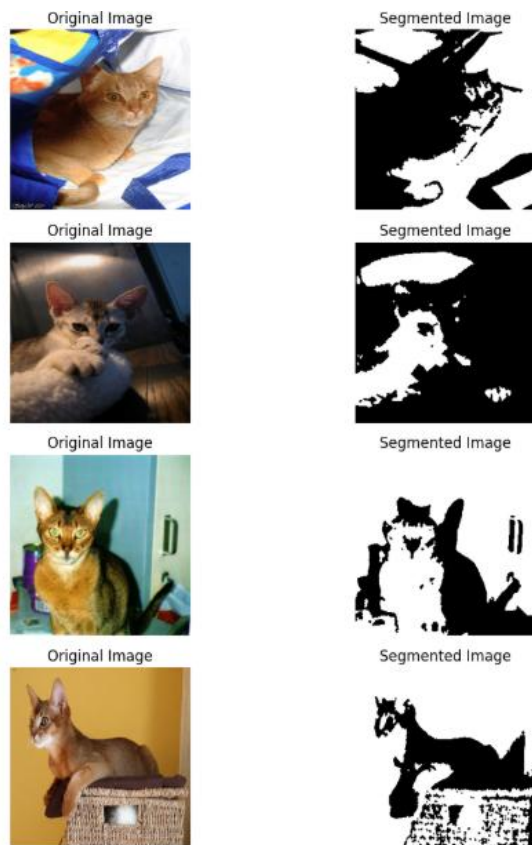


The comparison helps to understand how closely the predicted segmentation represents the actual pet region.

7.7 Multiple Image Testing

The segmentation process is tested on multiple images from the dataset. Otsu thresholding followed by opening and closing is applied to each image.

Multiple Image Segmentation Results

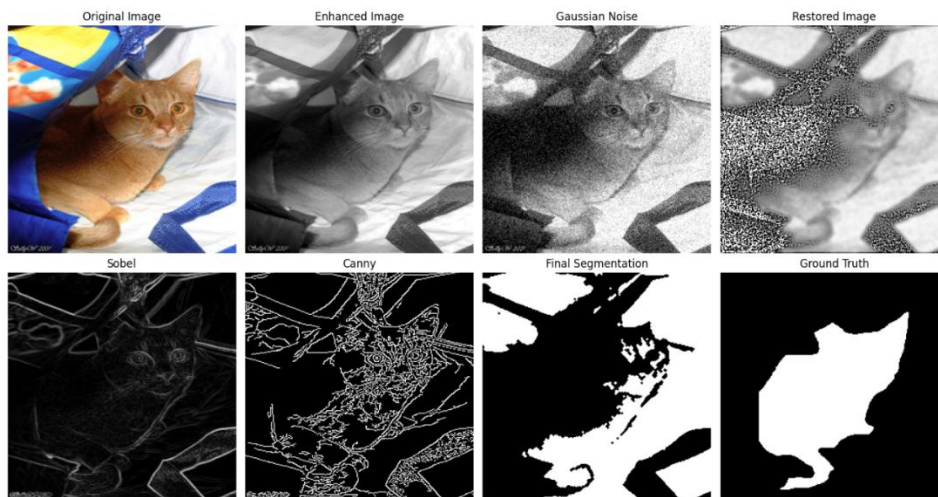


This testing shows how the segmentation technique performs on different pet images.

7.8 Final Project Results

The complete processing pipeline is displayed together, including the original image, enhanced image, noisy image, restored image, edge detection results, final segmentation, and ground truth.

Final Project Results



The final output provides an overall visual comparison of the different stages of the project.

8. Limitations

The project has the following limitations:

- Otsu thresholding may not work perfectly when the pet and background have similar intensity values.
- Edge detection may produce incomplete boundaries in some images.
- Restoration performance depends on the noise level and filter size.
- Traditional image processing techniques cannot recover details that are completely lost.
- Only a selected number of images are used for detailed testing.
- Performance may vary for images with complex backgrounds or low contrast.

9. Future Scope

The project can be improved in the future by:

- Using a larger and more diverse dataset.
- Applying adaptive filtering techniques.

- Using adaptive thresholding for difficult images.
- Exploring watershed segmentation.
- Applying frequency-domain filtering techniques.
- Using deep learning-based image restoration and segmentation.
- Evaluating the techniques on a larger number of images.
- Developing a real-time image processing application.

10. Conclusion

The project demonstrates a complete image processing workflow using the Oxford-IIIT Pet Dataset. Image enhancement techniques such as Gamma Transformation, Contrast Stretching, and Histogram Equalization are used to improve image appearance.

Gaussian and salt-and-pepper noise are added and removed using different filtering techniques. Sobel and Canny methods are used for edge detection, while Otsu thresholding is used for segmentation. Morphological operations such as erosion, dilation, opening, and closing are applied to improve the segmentation result.

MSE, PSNR, and SSIM are used to evaluate image restoration, while IoU, Dice Score, and Accuracy are used to evaluate segmentation. Overall, the project provides practical understanding of image enhancement, noise restoration, edge detection, thresholding, and morphological image processing.